

Physics of Cell Death

STABILITY LIMITS OF A CONFINED HYDROGEL SYSTEM

A cell can be understood as a confined hydrogel — a polymer network saturated with water and enclosed within an elastic membrane. In this framework, cell death is interpreted not primarily as a biochemical event, but as a mechanical transition: the system crossing beyond the boundaries of its own stability domain.

Core Idea

Setting aside molecular detail, a cell reduces to a physical object: a confined gel with elastic walls. Viability can then be framed as a problem in phase stability. As long as the system remains within a defined region of its state space, it persists. When driven beyond that region — by osmotic imbalance, structural degradation, or energy depletion — it fails.

At the level of physical description, this failure occurs in two dominant modes, which are distinguished mechanically rather than biologically.

State Variables

A minimal description requires five quantities:

W	Hydration — free water content within the network
P	Osmotic pressure — transmembrane differential
σ	Membrane tension — mechanical stress in the shell
C	Network connectivity — structural integrity of the polymer matrix
E	Available energy — capacity to maintain gradients and repair

These variables define the state space in which the cell exists as a trajectory.

The Stability Domain

There exists a bounded region in (W, P, σ , C, E) space satisfying three conditions:

- The membrane sustains mechanical load without rupture
- The polymer network retains connectivity
- The system remains bounded in volume and composition

This region corresponds to the viable state. Its boundary defines the stability limit.

The boundary is not perfectly sharp — it constitutes a probabilistic transition zone rather than a hard threshold.

Two Failure Modes

The geometry of the stability domain admits two dominant exit classes. These modes are topologically distinct and mechanically opposite in character.



Lytic Mode

SWELLING → RUPTURE

- Hydration (W) and osmotic pressure (P) increase
- Water influx drives volume expansion
- Membrane tension rises monotonically
- Rupture occurs at critical threshold

$$\sigma \geq \sigma_{\text{crit}}$$

Outcome: cytolysis — shell destruction

These modes are not alternative descriptions of the same process. One describes a reversible exit from the stability domain, the other an irreversible exit from the stability domain.

Syneresis-like Collapse

CONDENSATION → COLLAPSE

- Hydration (W) decreases or connectivity (C) increases
- Network contraction expels water
- Density and rigidity increase
- Molecular mobility diminishes

$W \rightarrow W_{\min} \quad \text{OR} \quad C \rightarrow C_{\max}$

Outcome: structural collapse of the internal matrix

... destroys the shell; the other destroys the internal structure. Both represent

FIG. 1 - TWO DOMINANT EXIT TRAJECTORIES FROM THE STABILITY DOMAIN (W- Σ PROJECTION)

The diagram illustrates the stability domain in the W - Σ projection. The vertical axis represents Membrane tension (Σ) and the horizontal axis represents Hydration (W). A dashed grey oval defines the stability domain. A green dot marks the 'Initial state' within this domain. Two trajectories exit from the initial state: a brown arrow pointing towards 'SYNERESIS' ($W \downarrow$, $C \uparrow$) and a red arrow pointing towards 'LYTIC' ($W \uparrow$, $\Sigma \uparrow$). Dashed lines indicate boundaries at $W=1$ and $\Sigma=1$.

The Role of Energy

Local deficits in available energy arise as a consequence of:

- This process is stochastic in both time and space. As available energy (E) declines:

- Energy depletion thus narrows the stability margin, increasing the probability of boundary crossing.

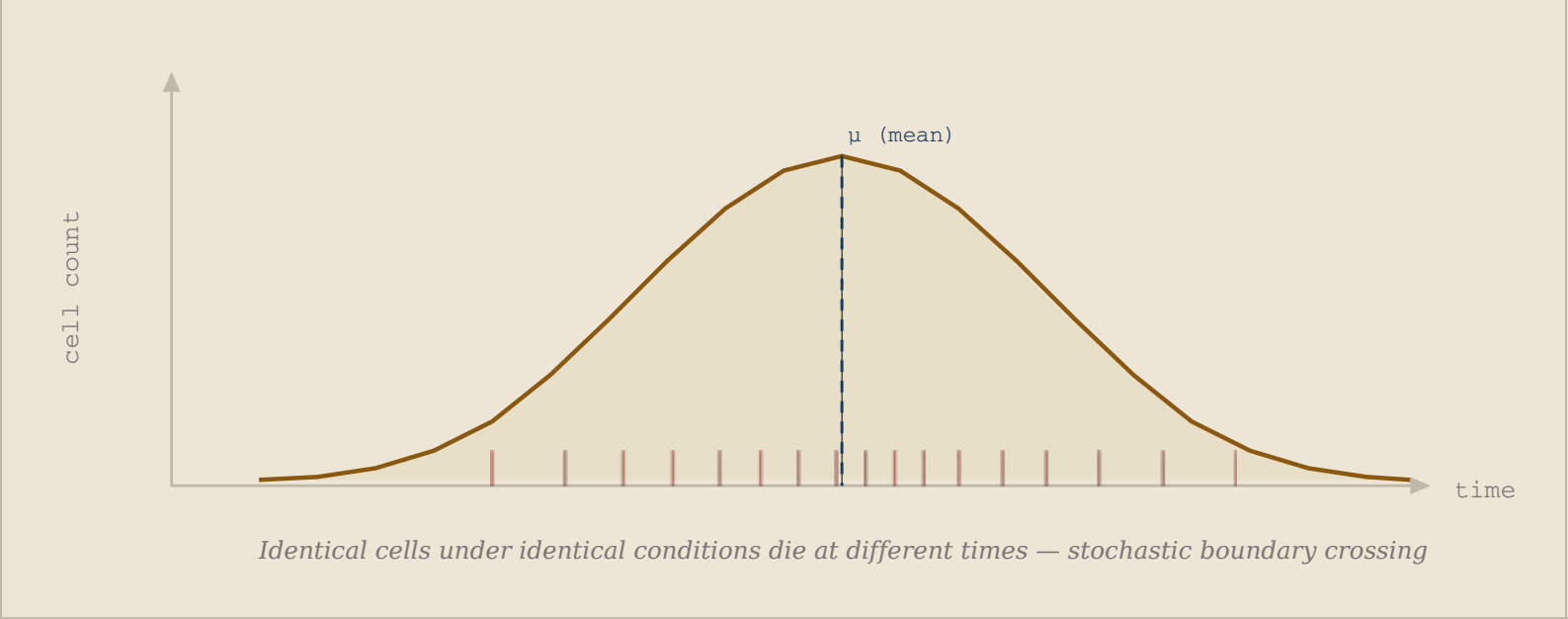
Stochasticity

Failure is not deterministic. Crossing the stability boundary occurs as a probabilistic process:

$$P_{\text{failure}} = f(W, P, \sigma, C, E)$$

This accounts for the variability in cell death timing across nominally identical populations: cells sharing the same conditions cross the boundary at different times. The stability limit is real; the moment of crossing is stochastic.

FIG. 2 — STOCHASTIC DISTRIBUTION OF CELL DEATH EVENTS IN AN ISOGENIC POPULATION (SCHEMATIC)



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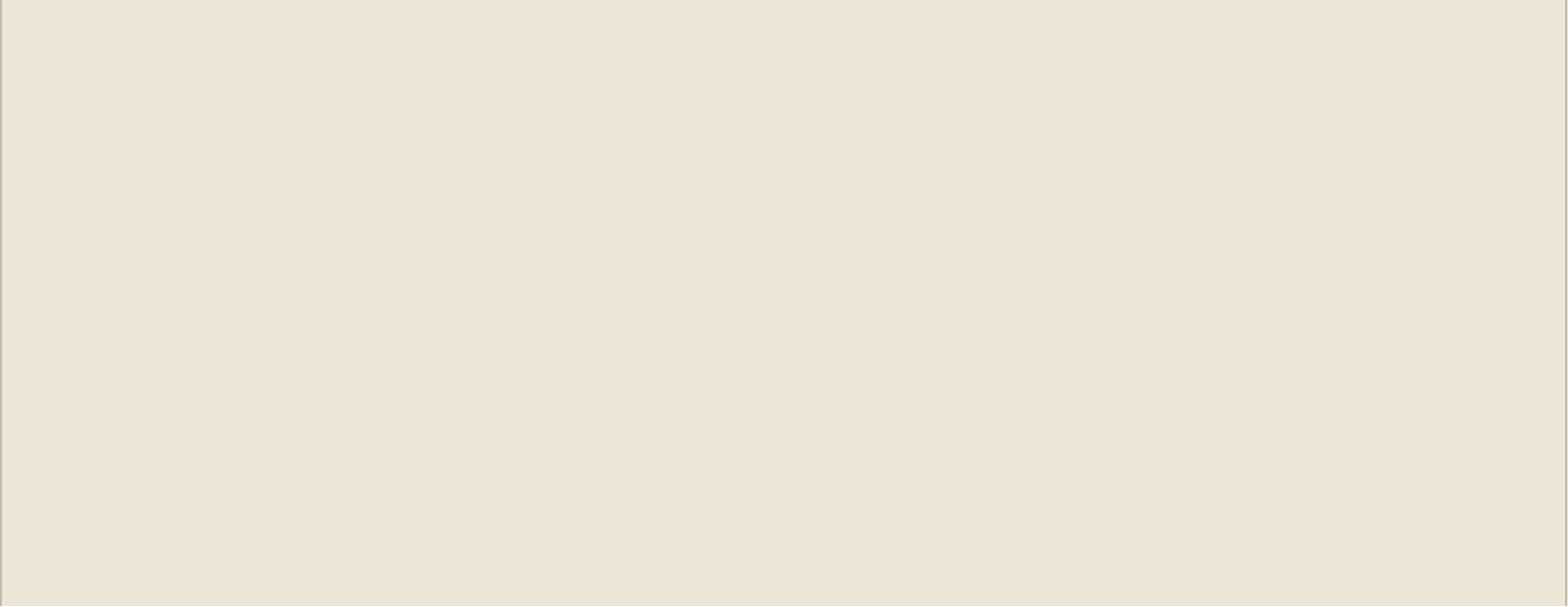
Phase Interpretation

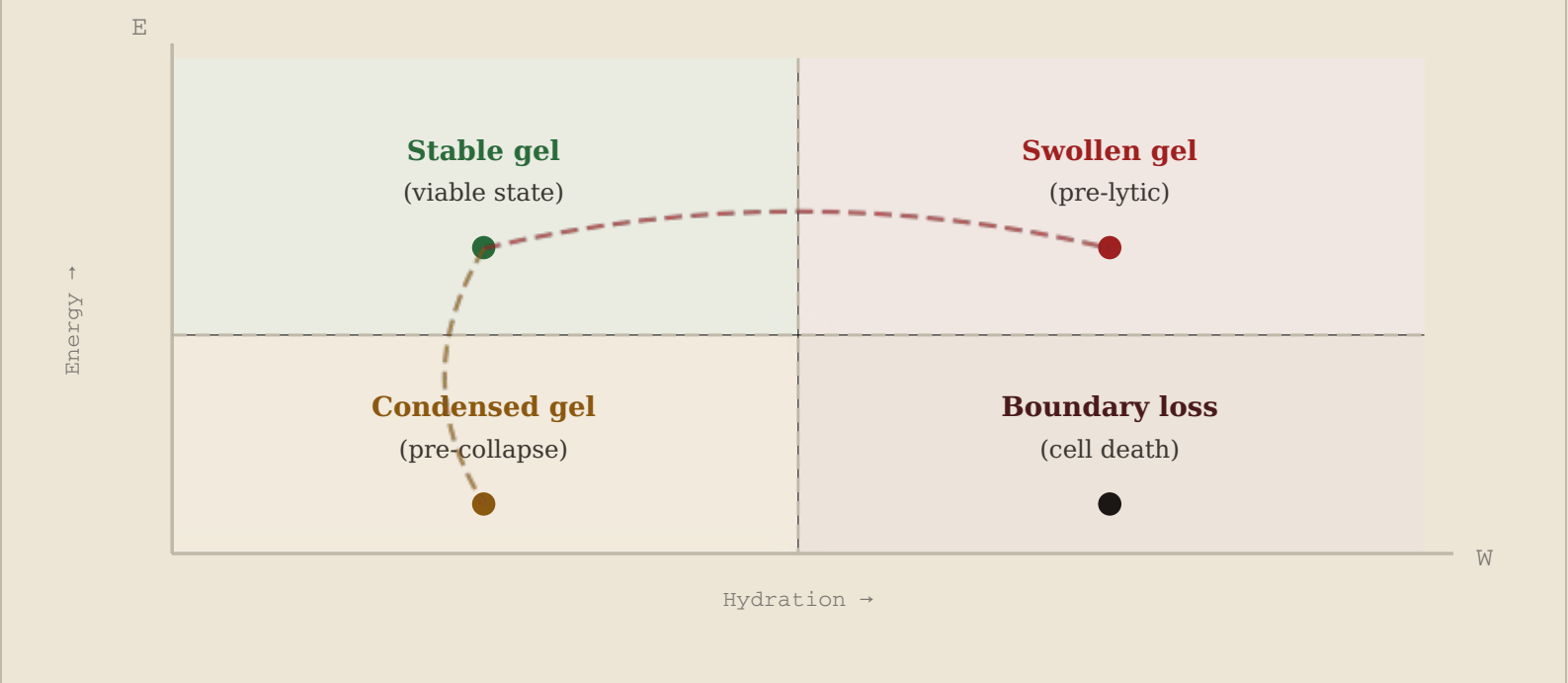
From the perspective of soft condensed matter physics, a cell behaves as a material capable of phase transitions. Four regimes can be identified:

PHASE	DESCRIPTION
● Stable gel	Viable state; system within stability domain
● Swollen gel	Pre-lytic state; tension rising toward σ_{crit}
● Condensed gel	Pre-collapse state; network contracting, $W \rightarrow W_{\text{min}}$
● Boundary loss	Death; irreversible exit from viable region

Cell death, in this framework, is the loss of phase stability in a confined soft matter system.

FIG. 3 — SCHEMATIC PHASE DIAGRAM: FOUR REGIMES IN (W, E) SPACE





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Why This Matters

This framework is universal in a restricted sense: it is independent of the specific molecular nature of the system and applies to any structure with analogous physical organization.

Traditional classifications of cell death — apoptosis, necrosis, pyroptosis, and others — describe mechanisms at the biological level. The present model provides the physical envelope within which those mechanisms are realized.

Any cell death pathway must manifest at the physical level through one of the described modes, regardless of its molecular implementation.

This approach has practical implications: interventions aimed at preventing or inducing cell death must act on at least one of the governing variables. Knowing which mode a system is approaching in advance determines the class of effective interventions.

SUMMARY

Not "apoptosis versus necrosis" — but two dominant physical failure modes of a confined soft matter system.

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